

Muhammad Hassan Khalid2IL-5692

(44) $A = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 2 & 5 \\ 6 & 8 & -1 \end{bmatrix} \rightarrow \begin{matrix} v_2 = (1, 1, 2, 8) \\ v_3 = (-5, 1, 5, -7) \end{matrix}$

$$\{v_1, v_2, v_3\} \xrightarrow{G-S} \{\hat{v}_1, \hat{v}_2, \hat{v}_3\}$$

$$\text{proj}_{v_1} v_2 = \frac{\langle v_2, v_1 \rangle v_1}{\|v_1\|^2}$$

$$\|v_1\|^2 = \langle v_1, v_1 \rangle = 80$$

$$\langle v_2, v_1 \rangle = 60$$

$$\{v_1\}; v_1 = v_1 = (6, 2, -2, 6)$$

$$\{v_1, v_2\} v_2 \rightarrow v_2 \text{ s.t. } v_2 \perp v_1$$

$$v_2 = v_2 - \text{proj}_{v_1} v_2$$

$$v_2 = (1, 1, 2, 8) - 3(6, 2, -2, 6)$$

$$v_2 = \frac{1}{2}(-7, -1, -1, 7)$$

$$\bullet \text{proj}_{v_1} v_3 = \frac{\langle v_3, v_1 \rangle v_1}{\|v_1\|^2}; \|v_1\|^2 = 80$$

$$\{v_1, v_2, v_3\} v_3 \rightarrow v_3 \text{ s.t. } v_3 \perp v_1$$

$$v_3 = v_3 - \text{proj}_{v_1} v_3 - \text{proj}_{v_2} v_3$$

$$\bullet \text{proj}_{v_2} v_3 = \frac{\langle v_3, v_2 \rangle v_2}{\|v_2\|^2}$$

$$v_3 = (-5, 1, 5, -7) + (6, 2, -2, 6)$$

$$\|v_1\|^2 = 25; \langle v_3, v_2 \rangle = -10$$

$$+ 2(5)(-7, -1, -1, 7)$$

$$\langle v_3, v_1 \rangle = -80$$

$$v_3 = \frac{1}{5}(-2, 14, 14, 2)$$

Orthogonal Set

$$\{v_1, v_2, v_3\} \xrightarrow{\text{Normal}} \{\hat{v}_1, \hat{v}_2, \hat{v}_3\}$$

$$v_1 = (6, 2, -2, 6) \rightarrow \hat{v}_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2\sqrt{5}}(3, 1, -1, 3)$$

$$v_2 = \frac{1}{2}(-7, -1, -1, 7) \rightarrow \hat{v}_2 = \frac{v_2}{\|v_2\|} = \frac{1}{10}(-7, -1, -1, 7)$$

$$v_3 = \frac{1}{5}(-2, 14, 14, 2) \rightarrow \hat{v}_3 = \frac{v_3}{\|v_3\|} = \frac{1}{10}(-1, 7, 7, 1)$$

$$O = (\hat{v}_1, \hat{v}_2, \hat{v}_3)$$